Hall Ticket Number:

MR-21

Code: B415CB4S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025/January 2026

Object Oriented Programming Using Java

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List any four principles of Object-Oriented Programming.	2	1	1	1
1. b)	Explain the difference between method overloading and overriding with an example.	2	2	1	1
1. c)	Define a package in Java. What are the steps to import a user-defined package?	2	2	2	1
1. d)	Illustrate the use of throws keyword in exception handling.	2	3	2	2
1. e)	What is the purpose of thread synchronization?	2	2	3	1
1. f)	Differentiate between byte stream and character stream.	2	2	3	2
1. g)	Define Iterator. How is it used in a Collection?	2	2	4	1
1. h)	Write the difference between HashMap and TreeMap.	2	2	4	1
1. i)	What is event delegation? Give one example.	2	2	5	2
1. j)	Define LayoutManager and mention any two of its types.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the structure of a Java program and the role of JVM and JDK.	5	2	1	1
2. b)	Write a Java program to demonstrate the use of "this" and "super" keywords.	5	2	1	1,2
	OR				
2. c)	Discuss access modifiers in Java with examples.	5	2	1	1
2. d)	Write a Java program to demonstrate single and multilevel inheritance.	5	2	1	1,2
3. a)	Explain how interfaces support multiple inheritance in Java.	5	2	2	1
3. b)	Develop a Java program implementing multiple interfaces for calculating area and volume.	5	3	2	1,2
	OR				
3. c)	Explain different types of exceptions in Java.	5	2	2	1
3. d)	Develop a Java program using try-catch-finally to handle ArrayIndexOutOfBoundsException.	5	3	2	1,2,3
4. a)	Explain the thread life cycle and how priorities affect scheduling.	5	2	3	1
4. b)	Write a Java program to create two threads and run them concurrently.	5	3	3	1,2
	OR				
4. c)	Explain file handling using FileReader and BufferedReader classes.	5	2	3	1
4. d)	Develop a Java program to count the number of words in a text file.	5	3	3	1,2
5. a)	Categorise the major interfaces in the Java Collections framework? Explain any three.	5	3	4	1
5. b)	Implement a Java program to demonstrate sorting using Collections.sort().	5	3	4	1,2

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	OR				
5. c)	Explain the role of HashSet, LinkedHashSet, and TreeSet with examples.	5	2	4	1
5. d)	Develop a Java program that counts unique words from a file using a HashSet.	5	4	4	1,2,3
6. a)	Discuss the architecture of Swing and its advantages over AWT.	5	2	5	1
6. b)	Develop a Java Swing program to display a calculator GUI.	5	4	5	1,2
	OR				
6. c)	Explain different layout managers used in GUI design.	5	2	5	1
6. d)	Develop a Java program to demonstrate mouse and keyboard event handling using adapter classes.	5	4	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome